Spread of Misinformation During Public Health Crises: A Network Approach

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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Bengaluru

2026

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April 2026

DECLARATION

We, hereby declare that the project report entitled **"Spread of Misinformation During Public Health Crises: A Network Approach"** is an original work done in the **Department of Computer Science and Systems Engineering, GITAM School of Computer Science and Engineering, GITAM (Deemed to be University), Bengaluru** submitted in partial fulfilment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree.

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ABSTRACT

The rapid spread of COVID-19-related misinformation during public health emergencies poses serious threats to healthcare systems, public safety, and societal trust. Misleading claims about cures, vaccines, treatments, and preventive measures can influence individual health decisions, increase vaccine hesitancy, and undermine evidence-based medical guidance. Social media platforms further accelerate the dissemination of such misinformation through text posts, shared URLs, and screenshots, making timely verification both complex and essential. This project addresses the challenge of detecting and verifying health-related claims in real time across multiple input formats, including text, URLs, and images, while ensuring explainability, reliability, and privacy preservation.

Existing misinformation detection systems often rely on keyword-based or text-only approaches that lack contextual understanding and fail to detect paraphrased or semantically similar claims. Many systems demonstrate monolingual bias, limited support for image-based verification, and insufficient transparency in their decision-making processes. Furthermore, black-box classification models provide limited interpretability, reducing user confidence in automated verification outcomes and limiting their practical usability during public health crises.

To overcome these limitations, this study proposes a multi-modal misinformation detection framework that integrates Sentence-BERT-based semantic similarity analysis for contextual meaning extraction, MinHash and Locality Sensitive Hashing-based claim similarity networks to identify repeated misinformation patterns, and rule-based intent and negation detection to prevent incorrect validation of strong medical claims such as cures or guaranteed treatments. The system incorporates a manually curated offline knowledge base constructed from authoritative sources including WHO, CDC, and NHS, thereby enabling privacy-preserving and trustworthy verification without reliance on external cloud services. Experimental evaluation demonstrates that the integration of semantic similarity modeling, network-based risk scoring, and intent-aware decision logic significantly improves reliability, transparency, explainability, and scalability, making the system a robust framework for detection of misinformation during public health crises.

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CHAPTER 1: INTRODUCTION

During public health emergencies such as the COVID-19 pandemic, misinformation spreads rapidly across social media through text messages, images, and shared URLs. False claims related to cures, vaccines, treatments, and preventive measures can undermine public trust, influence harmful health behaviors, and weaken confidence in scientific institutions. The widespread circulation of misleading information creates significant challenges for healthcare systems and policymakers managing crisis situations.

Existing misinformation detection systems have several limitations. Many rely on keyword-based filtering or traditional machine learning techniques that analyze only textual data, limiting their ability to capture contextual meaning and paraphrased claims. Many tools also lack multilingual support and mechanisms to verify image-based misinformation. Additionally, several systems depend on third-party APIs and operate as black-box models, offering limited transparency. These limitations highlight the need for an explainable, multi-modal, and privacy-preserving misinformation detection framework that integrates semantic analysis, claim similarity networks, and trusted health knowledge sources.

1.1 Purpose

The purpose of this project is to design and implement a multi-modal COVID-19 misinformation detection system capable of verifying claims in the form of text, URLs, or images. The system classifies claims as CORRECT, MISINFORMATION, or UNVERIFIED using Sentence-BERT semantic similarity, network-based risk scoring through MinHash and Locality Sensitive Hashing, intent-aware rule logic, and a locally maintained knowledge base constructed from trusted sources such as WHO, CDC, and NHS.

This project was developed in partial fulfillment of the requirements for the B.Tech Capstone Project under faculty supervision. The system is designed to operate offline while minimizing dependence on external cloud services and ensuring transparency, reproducibility, and scalability.

1.2 Scope

The scope of this project is limited to COVID-19-related health misinformation. The knowledge base is constructed using verified information from official health organizations to

ensure factual reliability. The system primarily operates in English but supports multilingual input through automatic language detection and translation.

Due to academic constraints, the system does not perform large-scale social media crawling or full social network analysis. Instead, it implements a lightweight claim similarity network using MinHash and LSH. The system focuses on claim verification rather than user behavior analysis or misinformation propagation modeling. Additionally, it does not retrain pre-trained models, instead emphasizing structured integration and controlled knowledge base expansion.

CHAPTER 2: LITERATURE SURVEY

The rapid spread of misinformation during the COVID-19 pandemic has attracted significant research attention across disciplines such as information systems, artificial intelligence, social network analysis, and crisis communication. This section reviews key studies related to social media misinformation, trust modeling, machine learning-based detection, and network-based misinformation analysis.

2.1 Social Media Trust and Shared Situational Awareness

Shahbazi and Bunker (2024) [1] examined how public health agencies used social media platforms such as Facebook to combat misinformation and promote Shared Situational Awareness (SSA) during the COVID-19 crisis. Using a netnographic approach and field interviews, the study analyzed crisis communication strategies based on three key constructs: information, communication, and trust.

The study found that social media can support rapid dissemination of accurate information, but ineffective communication strategies and declining public trust can weaken SSA formation. A key limitation identified is the limited focus on public-agency trust dynamics in prior SSA research.

2.2 Emotion and User Behavior-Based Misinformation Detection

Indu and Thampi (2024) [2] proposed a misinformation detection model combining emotion analysis and user behavior features. Their approach used Recurrent Neural Networks (RNNs) to extract emotional signals from tweets and applied a fuzzy inference system for classification.

Although the model performed well on Twitter rumor datasets, it was limited to textual content and profile-level attributes. The system also considered only four emotional states and lacked support for multimedia misinformation such as images or videos.

2.3 Machine Learning and Deep Learning for COVID-19 Fake News Detection

Hussna et al. (2024) [3] conducted a comprehensive review of COVID-19 misinformation detection research on X (formerly Twitter). The study analyzed 87 research papers and categorized detection techniques into traditional machine learning, deep learning, transformer-based models, and graph-based approaches.

The review found that most studies rely heavily on deep neural networks such as CNNs, LSTMs, and transformer models. While these methods achieve high accuracy, they often suffer from high computational cost, overfitting issues, and limited availability of large benchmark datasets.

2.4 Content Consistency and Behavioral Influence Analysis

Zhang et al. (2023) [4] investigated the relationship between misinformation content and public engagement behavior on Twitter. The study examined how consistency between tweets and user comments influences information sharing behavior using regression models and BERT-based cosine similarity.

The results showed that higher content consistency increases perceived credibility and sharing behavior. However, the study focused mainly on textual sentiment analysis and did not consider multimedia misinformation or long-term misinformation trends.

2.5 Network-Based Misinformation Dissemination Analysis

Röchert et al. (2021) [5] analyzed more than 10,000 YouTube videos and 2.58 million comments to study misinformation dissemination patterns. Using fine-tuned BERT models and network analysis techniques, the study examined informational homogeneity in video-comment networks.

The findings showed that misinformation does not exist in completely isolated echo chambers but often coexists with accurate information in interconnected networks. However, the study relied mainly on textual metadata and did not analyze multimedia misinformation signals.

2.6 Identified Research Gaps

From the reviewed literature [1]– [5], several limitations can be identified:

1. Most systems focus primarily on text-based misinformation detection and lack multi-modal support.
2. Heavy reliance on deep neural networks often results in black-box models with limited explainability.
3. Network-based studies rarely combine semantic similarity with risk modeling.
4. Limited support exists for multilingual misinformation detection.
5. Few systems incorporate intent-aware validation for distinguishing cure, treatment, and prevention claims.

6. Offline and privacy-preserving misinformation detection frameworks are rarely explored.

2.7 Motivation for the Proposed Work

Although existing studies demonstrate progress in semantic modeling and network analysis, an integrated framework combining semantic similarity, intent-aware validation, entity verification, and network-based risk scoring remains underexplored. Additionally, many systems lack transparency and offline capability.

Therefore, the proposed system addresses these gaps by:

- Supporting multi-modal inputs (text, URL, image)
- Integrating Sentence-BERT semantic similarity
- Applying intent-aware and negation-aware validation
- Using MinHash and LSH for claim similarity network modelling
- Incorporating entity validation through offline structured data
- Providing transparent rule-based decision logic
- Maintaining offline and privacy-preserving operation

CHAPTER 3: PROBLEM STATEMENT

The rapid spread of COVID-19 health misinformation during public health crises has significantly impacted healthcare systems, government responses, and public trust in scientific institutions. False claims related to cures, vaccines, preventive measures, and transmission circulate widely on social media through text messages, URLs, screenshots, and visual content. Such misinformation can influence health decisions, increase vaccine hesitancy, generate public panic, and weaken evidence-based medical guidance. The large-scale dissemination of misleading information makes real-time verification both complex and essential.

Existing misinformation detection systems face several technical limitations. Many approaches rely on keyword-based filtering or text-only natural language processing methods that fail to capture contextual meaning or detect paraphrased claims. These systems often cannot verify misinformation presented through images or URLs and struggle with multilingual content. Additionally, manual fact-checking methods lack scalability for real-time scenarios. Many automated classification models operate as black-box systems with limited transparency, reducing user trust. Dependence on external APIs and online services also raises concerns related to privacy, reliability, and long-term sustainability.

These limitations highlight the need for a multi-modal, explainable, scalable, and privacy-preserving misinformation detection framework capable of supporting reliable verification during public health emergencies.

3.1 Objectives

The primary objectives of this project are:

- Develop a multi-modal misinformation detection system capable of verifying claims from text, URLs, and images using trusted health sources.
- Design an explainable fact-checking pipeline integrating semantic similarity, claim similarity networks, and threshold-based decision logic to classify claims as CORRECT, MISINFORMATION, or UNVERIFIED with supporting evidence and confidence scores.
- Build a locally curated knowledge base using verified information from WHO, CDC, and NHS to enable offline and privacy-preserving validation.
- Support multilingual misinformation detection through automatic language detection and translation.

- Improve detection beyond keyword-based methods by using Sentence-BERT embeddings, MinHash, and Locality Sensitive Hashing for scalable and context-aware claim analysis.

CHAPTER 4: SOFTWARE AND HARDWARE SPECIFICATIONS

4.1 Introduction

This chapter describes the software and hardware requirements for developing and deploying the proposed multi-modal COVID-19 misinformation detection system. The system is implemented in Python and integrates web technologies, natural language processing models, semantic similarity techniques, and network-based claim analysis. The design emphasizes offline operation, privacy preservation, modular architecture, and efficient performance.

4.2 Specific Requirements

4.2.1 Functional Requirements

The system accepts user inputs in multiple formats, including text, URLs, and images containing textual information. It extracts and preprocesses the content and performs semantic similarity analysis using Sentence-BERT embeddings. Input claims are compared with verified facts and known misinformation patterns stored in a locally maintained knowledge base.

The system incorporates intent-aware validation and negation detection to prevent incorrect classification of medical claims. It also computes a network risk score using MinHash and Locality Sensitive Hashing (LSH) to detect repeated misinformation patterns.

The output includes a classification label (CORRECT, MISINFORMATION, or UNVERIFIED), supporting evidence, confidence score, and explanation. The system also supports multilingual inputs through language detection and translation. Optional local LLM reasoning improves interpretability without relying on external APIs.

4.2.2 Non-Functional Requirements

The system ensures privacy preservation by operating primarily in offline mode without dependence on external cloud services. It provides transparent and interpretable outputs supported by evidence-based explanations.

The architecture is modular to allow future scalability and knowledge base expansion. Performance requirements include efficient semantic similarity computation and fast near-

duplicate detection through LSH indexing. The system is designed to be reliable, maintainable, and reproducible within academic computational constraints.

4.3 Hardware and Software Requirements

4.3.1 Hardware Requirements

The system is optimized for moderate hardware configurations and does not require specialized computing resources. Recommended specifications include:

- **Processor:** Intel Core i5 or equivalent
- **RAM:** Minimum 8 GB
- **Storage:** 256 GB SSD
- **GPU:** Not required (CPU-based execution supported)
- **Operating System:** Windows 10 / Linux

The lightweight embedding model and efficient indexing methods allow smooth execution without GPU acceleration.

4.3.2 Software Requirements

The system is developed using **Python 3.x** and integrates several libraries to support multi-modal verification and semantic analysis.

Primary software components include:

- **Streamlit** – Web interface
- **Sentence-BERT (all-MiniLM-L6-v2)** – Semantic embeddings
- **Ollama (Gemma2:2B)** – Local LLM reasoning
- **langdetect** – Language detection
- **deep-translator** – Multilingual translation
- **EasyOCR** – Image text extraction
- **spaCy** – Named entity recognition
- **datasketch** – MinHash and LSH implementation
- **JSON files** – Offline data storage

All dependencies are managed through a **requirements.txt** file to ensure reproducibility and consistent system setup.

CHAPTER 5: PROPOSED SYSTEM

To address the challenges identified in the problem statement, this project proposes a multi-modal and network-aware COVID-19 misinformation detection system. The system verifies health-related claims submitted as text, URLs, or images and classifies them as CORRECT, MISINFORMATION, or UNVERIFIED.

The framework integrates semantic similarity analysis, intent-aware validation, and network-based claim similarity modeling to ensure reliable and explainable verification while supporting offline operation and privacy preservation.

5.1 System Overview

The proposed system follows a structured verification pipeline, as illustrated in Figure 5.1.1. Each stage processes the input claim and passes the result to the next stage for further analysis.

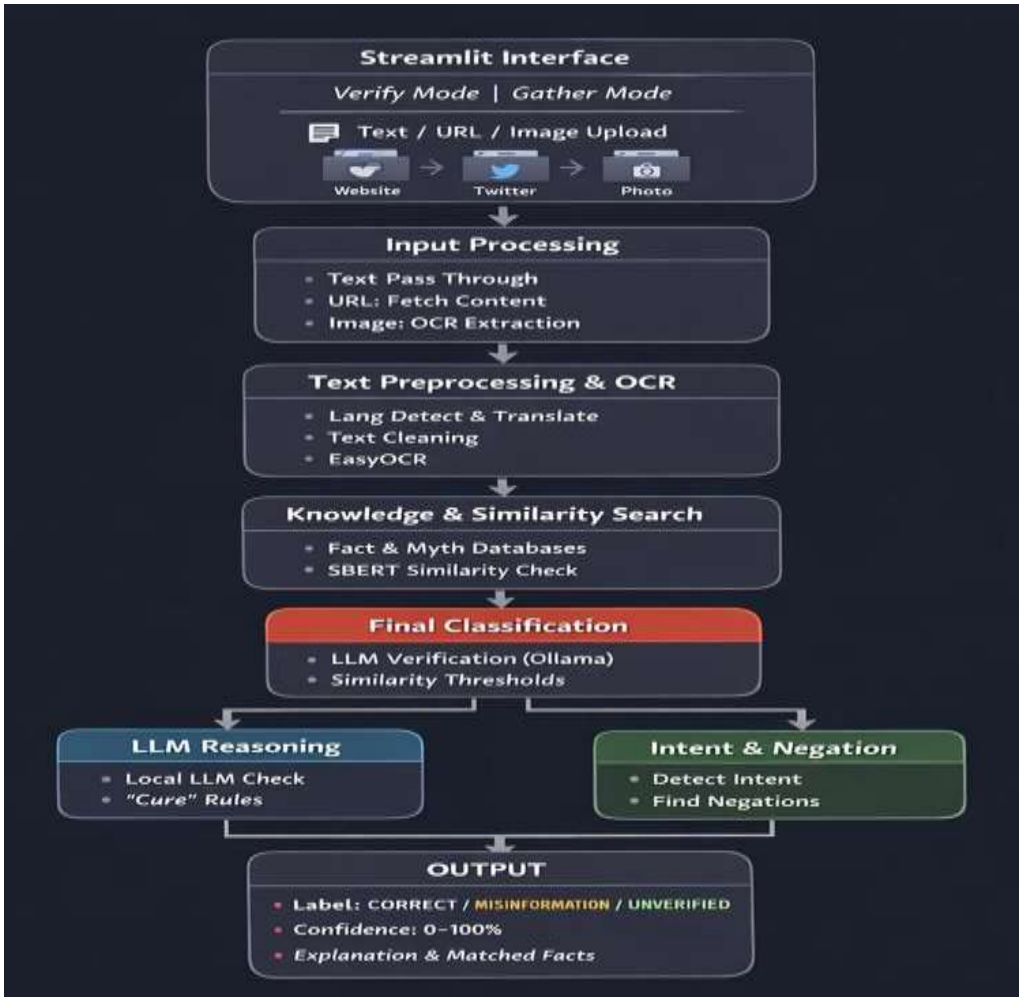

Figure 5.1.1: Workflow of the Proposed Multi-Modal Misinformation Detection System

5.1.1 Streamlit Interface and Input Handling

The system provides a Streamlit-based web interface where users can submit claims through text input, URLs, or image uploads.

- Text inputs are passed directly to the processing pipeline.
- URL inputs are used to extract relevant webpage content.
- Image inputs are processed using Optical Character Recognition (OCR) to extract embedded text.

All inputs are converted into standardized textual format before further analysis.

5.1.2 Input Processing

The extracted claim is prepared for analysis through basic processing steps:

- Text input is directly forwarded.
- URL content is fetched and cleaned.
- Images are converted into text using EasyOCR.

This ensures that claims from different formats are converted into a common textual representation.

5.1.3 Text Preprocessing

The extracted text undergoes preprocessing to improve analysis accuracy.

This stage includes:

- Language detection and translation of non-English inputs
- Text normalization, including lowercasing and removal of special characters
- Cleaning and formatting of the extracted text

These steps produce a clean and consistent input for semantic analysis.

5.1.4 Knowledge Base and Similarity Search

The system compares the processed claim with entries stored in a locally maintained knowledge base.

The knowledge base contains:

- Verified facts

- Known misinformation patterns

The claim is encoded using Sentence-BERT embeddings, and cosine similarity is used to identify semantically similar entries.

5.1.5 Final Classification Process

The classification stage integrates several analysis components to determine the result.

LLM Reasoning

A locally deployed LLM through Ollama can generate human-readable explanations based on matched evidence.

Intent and Negation Detection

A rule-based module analyzes the claim to detect:

- The type of medical claim (prevention, treatment, cure)
- Negation patterns such as “does not cure” or “not effective”

This prevents incorrect validation of strong medical claims.

5.1.6 Network-Based Risk Analysis

The system maintains a similarity-based claim network using MinHash and Locality Sensitive Hashing (LSH). When a new claim is processed, the system checks whether similar claims were previously labeled as misinformation. A network risk score is calculated based on the proportion of similar misinformation claims.

5.1.7 Decision Engine and Output

The final decision is produced by combining:

- Semantic similarity scores
- Intent validation results
- Network risk scores
- Optional LLM reasoning

The system outputs:

- Classification label: CORRECT / MISINFORMATION / UNVERIFIED

- Confidence score (0–100%)
- Explanation and matched evidence

5.2 Key Features of the Proposed System

- Multi-modal input support (text, URL, image)
- Semantic similarity-based contextual verification
- Intent-aware and negation-aware validation
- Similarity-based claim network modelling
- Offline and privacy-preserving architecture
- Explainable outputs with confidence scores
- Controlled knowledge base expansion

5.3 Advantages Over Existing Systems

The proposed system improves traditional misinformation detection approaches by moving beyond keyword-based filtering and incorporating contextual semantic understanding.

The framework enhances interpretability through rule-based validation and evidence-backed explanations. It also supports multilingual inputs, offline operation, and network-based risk analysis, enabling reliable detection of clustered misinformation patterns.

Overall, the system provides a scalable, explainable, and privacy-preserving solution for detecting COVID-19 misinformation during public health crises.

CHAPTER 6: DESIGN

The proposed system follows a modular and layered architecture designed to ensure scalability, explainability, and offline operation. Each module performs a specific function within the misinformation verification pipeline. The architecture integrates semantic similarity analysis, intent-aware validation, entity verification, network-based similarity modeling, and LLM reasoning within a unified decision framework.

The overall workflow of the system is illustrated in Figure 6.1 and follows this sequence:

User Input → Input Handling → Preprocessing → Knowledge Base → Semantic Similarity → Intent Detection → Entity Validation → Network Risk Analysis → LLM Verification → Decision Engine → Final Output

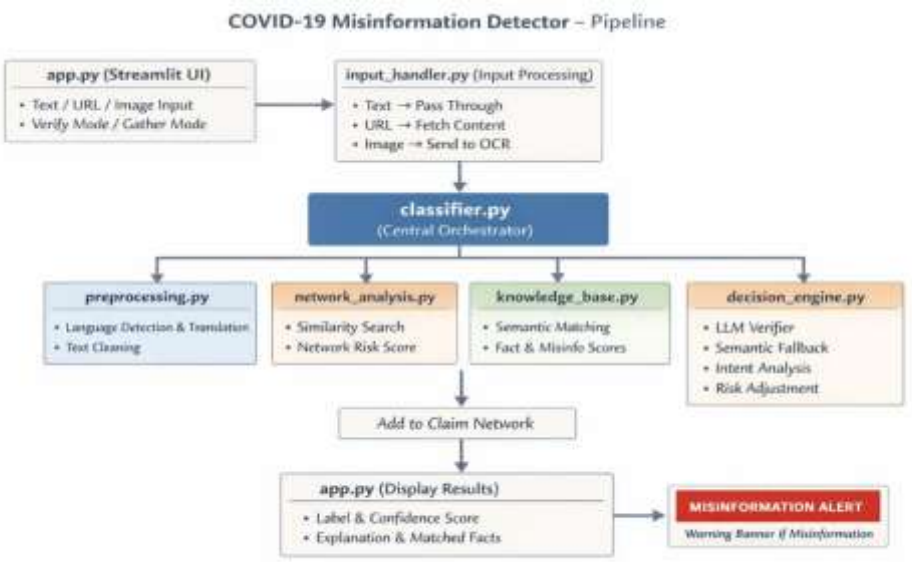

Figure 6.1: System Architecture

6.1 Architectural Layers

The architecture consists of four primary layers.

User Interface Layer

The system provides a Streamlit-based web interface where users can submit claims through text input, URLs, or image uploads.

The interface operates in two modes:

- **Verify Mode** – validates user-submitted claims
- **Gather Mode** – supports controlled knowledge base expansion

The system displays the classification label, confidence score, detected intent level, explanation with matched evidence, and network risk indicator. If misinformation is detected, a warning message is displayed.

Processing Layer

The processing layer performs the core claim analysis operations. It includes:

- Input handling
- Text preprocessing
- Semantic encoding
- Intent detection
- Entity validation
- Network similarity analysis

Each module processes the claim independently and contributes to the final decision.

Knowledge Layer

The system uses a two-tier knowledge base.

Tier 1: Verified facts and known misinformation patterns from WHO, CDC, and NHS. **Tier 2:** Structured entity information derived from offline Wikidata for validating numerical or factual claims.

Sentence-BERT embeddings for knowledge entries are precomputed and cached to improve similarity search performance.

Decision Layer

The decision engine integrates outputs from semantic similarity, intent validation, entity verification, network risk scoring, and LLM reasoning. Using predefined thresholds and weighted logic, the system assigns the final classification label and confidence score.

6.2 Methodology

The system follows a multi-stage verification pipeline, where each stage applies specific algorithms to analyze the input claim and determine its credibility.

6.2.1 Preprocessing

The preprocessing stage standardizes all input claims.

- Text input is directly processed.
- URLs are scraped to extract relevant textual content.
- Images are converted to text using EasyOCR.

Language detection is performed using langdetect, and non-English text is translated using deep-translator. Text normalization removes URLs, punctuation, and special characters to produce a clean input representation.

6.2.2 Knowledge Base Construction

The knowledge base is stored locally in JSON format and contains verified facts and known misinformation patterns collected from trusted sources. Each knowledge entry is encoded using Sentence-BERT (all-MiniLM-L6-v2) to generate semantic embeddings. These embeddings are precomputed and cached to enable efficient similarity comparison.

6.2.3 Semantic Similarity (Sentence-BERT + Cosine Similarity)

The processed claim is encoded using the Sentence-BERT embedding model. Similarity between the claim and knowledge entries is computed using Cosine Similarity:

$$\text{Similarity} = (A \cdot B) / (\|A\| \times \|B\|)$$

Top-k matching results are retrieved based on similarity scores, enabling detection of paraphrased or semantically similar misinformation claims.

6.2.4 Intent Detection

Intent detection uses rule-based classification to identify the type of medical claim, such as:

- General wellness
- Symptom management
- Prevention
- Treatment
- Cure

Negation detection identifies phrases such as *"does not cure"* or *"cannot prevent"* to avoid incorrect classification.

6.2.5 Entity Validation

Entity validation ensures factual consistency beyond semantic similarity. Named entities such as vaccine names, drug names, organizations, and numerical values are extracted using spaCy Named Entity Recognition (NER). These entities are compared with structured entries from the offline Wikidata dataset to detect factual inconsistencies.

6.2.6 Network Risk Scoring (MinHash + LSH)

To identify repeated misinformation patterns, the system builds a similarity-based claim network. Each claim is converted into shingles and processed using MinHash, which approximates Jaccard similarity. Locality Sensitive Hashing (LSH) indexes these signatures to retrieve similar claims efficiently.

The neighborhood risk score is calculated as:

Risk = (Number of Misinformation Neighbors) / (Total Similar Neighbors)

If the risk exceeds predefined thresholds, the classification confidence is adjusted accordingly.

6.2.7 LLM Verification

A local LLM executed through Ollama (Gemma2:2B) generates structured explanations for the classification result. The LLM enhances interpretability but does not override rule-based constraints or entity validation results.

6.2.8 Decision Engine Logic

The final classification is determined by integrating:

- Semantic similarity score
- Intent validation results
- Entity validation outcome
- Network risk score
- Optional LLM reasoning

Based on predefined thresholds, the system assigns one of three labels: Correct, Misinformation and Unverified

The system outputs the classification label, confidence score, and supporting explanation, ensuring transparent and reproducible decision-making.

CHAPTER 7: TECHNOLOGIES AND ALGORITHMS

This chapter describes the technologies and algorithms used to develop the proposed multi-modal COVID-19 misinformation detection system. The system integrates web technologies, natural language processing models, semantic embedding methods, and similarity-based network algorithms to ensure efficient and explainable claim verification.

7.1 Technologies

The main technologies used in the system are summarized in Table 7.1.1.

Table 7.1.1: System Components and Technologies

Component	Technology Used
Web Interface	Streamlit
Text Embeddings	Sentence-BERT (all-MiniLM-L6-v2)
Local LLM Reasoning	Ollama (Gemma2:2B)
Language Detection	langdetect
Translation	deep-translator (Google Translate)
OCR (Image Text Extraction)	EasyOCR
Entity Recognition	spaCy
Near-Duplicate Detection	MinHash + LSH (datasketch)
Data Storage	JSON Files

Web Interface – Streamlit

Streamlit is used to build the web-based interface that allows users to submit text, URL, or image inputs. It displays verification results including classification labels, confidence scores, and explanations.

Text Embeddings – Sentence-BERT

Sentence-BERT (all-MiniLM-L6-v2) converts textual claims into semantic vector embeddings, enabling detection of paraphrased or contextually similar misinformation.

Local LLM Reasoning – Ollama

Ollama is used to run lightweight local LLM models such as Gemma2:2B, which generate human-readable explanations while maintaining offline operation.

Language Processing Tools

The system uses langdetect for language detection and deep-translator for translating non-English text into English before verification.

OCR – EasyOCR

EasyOCR extracts text from uploaded images such as screenshots or posters so that image-based claims can be analyzed using the same verification pipeline.

Entity Recognition – spaCy

spaCy is used for Named Entity Recognition (NER) to detect entities such as vaccine names, drug names, organizations, and numerical values.

Near-Duplicate Detection – MinHash + LSH

The datasketch library implements MinHash and Locality Sensitive Hashing (LSH) to efficiently identify similar or repeated misinformation claims.

Data Storage – JSON Files

Knowledge base entries, embeddings, and processed claims are stored locally in JSON format, enabling lightweight storage and offline functionality.

7.2 Algorithms

The algorithms used in the system are summarized in Table 7.2.1.

Table 7.2.1: Algorithms

Category	Algorithm	Module Applied
Semantic Modeling	Sentence-BERT (all-MiniLM-L6-v2)	Semantic Similarity Module
Similarity Measurement	Cosine Similarity	Semantic Similarity Module

Classification Logic	Threshold-Based Classification	Decision Engine
Network Modeling	MinHash (Minimum Hashing)	Network Analysis Module
Network Modeling	Locality Sensitive Hashing (LSH)	Network Analysis Module
Risk Evaluation	Neighborhood Risk Scoring	Network Analysis Module
Image Processing	EasyOCR	Input Handling / Preprocessing
Language Processing	Statistical Language Detection (langdetect)	Preprocessing Module
Translation	Neural Machine Translation (deep-translator)	Preprocessing Module
Text Cleaning	Regex-Based Text Normalization	Preprocessing Module
Entity Extraction	spaCy Named Entity Recognition (NER)	Entity Validation Module
Intent Classification	Rule-Based Intent Detection	Intent Detection Module

7.2.1 Core Classification Algorithms

The system primarily relies on semantic similarity and rule-based decision logic for claim verification.

- **Sentence-BERT** generates contextual embeddings for input claims.
- **Cosine Similarity** measures similarity between claim embeddings and knowledge base entries.
- **Threshold-based classification** assigns labels (CORRECT, MISINFORMATION, UNVERIFIED) based on similarity scores, intent validation, entity validation, and network risk analysis.
- **Local LLM Reasoning** enhances interpretability by providing structured

explanations but does not override rule-based constraints.

7.2.2 Network Algorithms

The system uses **MinHash and Locality Sensitive Hashing (LSH)** to build a similarity-based claim network. These algorithms enable efficient detection of near-duplicate claims and repeated misinformation patterns without constructing a full social network graph. A neighborhood risk score is computed based on the proportion of similar claims previously labeled as misinformation.

7.2.3 Text and Image Processing Algorithms

Several algorithms are used to standardize input claims before verification:

- **EasyOCR** extracts text from images.
- **langdetect** identifies the language of input text.
- **deep-translator** converts non-English text into English.
- **Regex-based normalization** cleans the text.
- **spaCy NER** extracts structured entities for validation.
- **Rule-based intent detection** identifies the level of medical claims.
- **Negation detection** prevents misclassification of corrective statements.

CHAPTER 8: IMPLEMENTATION

This chapter describes the implementation of the proposed multi-modal COVID-19 misinformation detection system. The system is developed using Python and follows a modular architecture, where each module performs a specific task in the verification pipeline. The implementation integrates semantic similarity modeling, rule-based validation, network analysis, and LLM reasoning while maintaining offline capability and extensibility.

8.1 Development Environment

The system is implemented using Python 3.x and supports both Windows and Linux operating systems. Development was performed using Visual Studio Code as the primary development environment.

The web interface is implemented using Streamlit, while optional local large language model execution is handled through Ollama.

The application is launched using:

```
streamlit run app.py
```

This command initializes all system modules and launches the web interface.

8.2 Project Directory Structure

The project follows a structured directory organization as illustrated in Figure 8.2.1

```
Project/
│
├── app.py
├── classifier.py
├── decision_engine.py
├── preprocessing.py
├── knowledge_base.py
├── intent_detector.py
├── entity_validator.py
├── network_analysis.py
├── llm_verifier.py
```

```
├── input_handler.py
├── evaluation.py
├── fact_gatherer.py
├── knowledge_ingestion.py
├── tester_feedback.py
├── config.py
│
├── requirements.txt
├── README.md
│
├── data/
│   ├── verified_facts.json
│   ├── known_misinformation.json
│   ├── wikidata_entities.json
│   ├── staged_claims.json
│   ├── test_cases.json
│   └── feedback_log.json
│
├── cache/
│   ├── embeddings_cache.pkl
│   ├── verified_embeddings.pkl
│   ├── misinformation_embeddings.pkl
│   └── wikidata_cache.pkl
│
├── network_storage/
│   ├── lsh_index.pkl
│   ├── minhash_signatures.pkl
│   └── claim_graph.pkl
│
├── logs/
│   ├── system_logs.txt
│   ├── evaluation_results.json
```

```
│   └── error_logs.txt
│
├── models/
│   ├── sbert_model/
│   └── ollama_config.json
│
└── static/
├── uploaded_images/
└── temporary_files/
```

Figure 8.2.1: Project Directory Structure

Root Directory

Contains the main Python modules responsible for system execution and orchestration.

data/

Stores structured datasets and knowledge resources.

- verified_facts.json – Verified health information
- known_misinformation.json – Known misinformation patterns
- wikidata_entities.json – Structured entity attributes
- staged_claims.json – Newly collected claims awaiting validation
- test_cases.json – Evaluation dataset
- feedback_log.json – Tester feedback records

cache/

Stores cached resources to improve runtime performance.

- Precomputed Sentence-BERT embeddings
- Cached entity mappings
- Avoids repeated computation

network_storage/

Stores artifacts related to the similarity-based claim network.

- MinHash signatures

- LSH index
- Persisted similarity graph

logs/

Stores execution logs, evaluation outputs, and error traces.

models/

Stores model configuration files for SBERT and Ollama runtime.

static/

Temporary storage for uploaded images and intermediate files.

8.3 Main System Modules

The system is implemented using modular Python files, where each module performs a specific function in the pipeline.

Table 8.3.1: Primary Files and Responsibilities

File Name	Responsibility
app.py	Streamlit user interface and system orchestration
classifier.py	Core semantic similarity computation pipeline
decision_engine.py	Final label assignment logic
preprocessing.py	Text cleaning, language detection, translation, OCR integration
knowledge_base.py	Loading and managing verified facts and misinformation patterns
intent_detector.py	Rule-based intent classification and negation detection
entity_validator.py	Named entity extraction and Wikidata validation
network_analysis.py	Claim similarity network using MinHash and LSH
llm_verifier.py	Local LLM integration via Ollama
input_handler.py	Processing text, URL, and image inputs
evaluation.py	Controlled scraping from trusted sources

fact_gatherer.py	Controlled scraping from trusted sources
knowledge_ingestion.py	Offline knowledge expansion mechanism

8.4 Implementation Workflow

The implementation follows a structured pipeline where modules process the input sequentially.

8.4.1 User Interface (app.py)

The Streamlit interface provides two modes:

- **Verify Mode** – verifies misinformation claims
- **Gather Mode** – supports controlled knowledge collection

Users can submit text, URLs, or images. The interface displays:

- Classification label
- Confidence score
- Matched evidence
- Intent level
- Network risk score
- LLM explanation

If misinformation is detected, a warning message is displayed.

8.4.2 Input Handling (input_handler.py)

This module determines the input type and processes it accordingly.

Text input → passed directly to preprocessing
URL input → webpage content extraction
Image input → text extraction using EasyOCR

All inputs are converted into standardized textual format.

8.4.3 Text Preprocessing (preprocessing.py)

The preprocessing module standardizes input text.

Processing steps include:

- Language detection using **langdetect**
- Translation of non-English text using **deep-translator**
- Text normalization (lowercasing, punctuation removal, whitespace cleaning)
- OCR integration for image inputs

This ensures consistent input for semantic analysis.

8.4.4 Knowledge Base Management (knowledge_base.py)

The knowledge base is stored in local JSON files and organized into two tiers.

Tier 1: Verified facts and known misinformation patterns
Tier 2: Structured entity data from Wikidata

Sentence-BERT embeddings for Tier 1 entries are precomputed and cached to improve performance.

8.4.5 Semantic Similarity (classifier.py)

The classifier module encodes the input claim using Sentence-BERT embeddings. Similarity between the claim and knowledge base entries is computed using Cosine Similarity. The top-k matching entries and their similarity scores are forwarded to the decision engine.

8.4.6 Intent Detection (intent_detector.py)

The intent detection module categorizes claims into medical intent levels.

Categories include:

- General wellness
- Symptom management
- Prevention
- Treatment
- Cure

Negation phrases such as *"does not cure"* are detected using pattern matching to prevent incorrect classification.

8.4.7 Entity Validation (entity_validator.py)

Named entities are extracted using spaCy NER. Entities such as vaccine names, drug names, organizations, and numerical values are validated against offline Wikidata data. If conflicts are detected, a validation flag is generated for the decision engine.

8.4.8 Network Analysis (network_analysis.py)

The similarity-based claim network is implemented using MinHash and Locality Sensitive Hashing (LSH).

Steps include:

- Generate MinHash signatures for claims
- Use LSH to retrieve similar claims efficiently
- Compute a neighborhood risk score based on similar misinformation claims

The similarity index is stored using pickle for persistence.

8.4.9 LLM Verification (llm_verifier.py)

This module connects to a locally running Ollama server. It generates structured explanations for classification results. However, LLM output does not override rule-based validation or entity checks.

8.4.10 Decision Engine (decision_engine.py)

The decision engine integrates outputs from:

- Semantic similarity scores
- Intent detection results
- Entity validation flags
- Network risk scores
- LLM reasoning

Based on predefined thresholds, the system assigns the final label:

CORRECT / MISINFORMATION / UNVERIFIED

8.5 Evaluation Framework

The evaluation.py module uses curated test cases to compute metrics such as confusion matrices and classification accuracy.

8.6 Knowledge Expansion and Feedback

The system supports controlled knowledge expansion through:

- **fact_gatherer.py** – collects information from trusted sources
- **knowledge_ingestion.py** – integrates new data into the knowledge base
- **tester_feedback.py** – logs feedback from authorized testers

8.7 System Configuration

The config.py file stores system parameters including:

- Similarity thresholds
- Decision weights
- Label definitions
- Directory paths

Centralized configuration improves maintainability and experimentation.

CHAPTER 9: EXPERIMENTAL SETUP

This chapter describes the experimental configuration used to evaluate the proposed multi-modal COVID-19 misinformation detection system. The experiments assess the effectiveness of semantic similarity modeling, intent-aware validation, entity verification, and network-based risk scoring for classifying health-related claims.

9.1 Hardware Configuration

The system was tested on a standard computing environment without requiring specialized hardware.

- **Processor:** Intel Core i5 (or equivalent)
- **RAM:** 8 GB
- **Storage:** 256 GB SSD
- **Operating System:** Windows 10 / Linux
- **GPU:** Not required (CPU-based execution)

The lightweight Sentence-BERT model and efficient LSH indexing enable smooth execution on CPU-based systems.

9.2 Software Environment

The experiments were conducted using the following software stack:

- Python 3.x
- Streamlit (Web Interface)
- Sentence-BERT (all-MiniLM-L6-v2)
- Ollama (Local LLM Runtime)
- spaCy (NER)
- EasyOCR
- Langdetect
- deep-translator
- datasketch (MinHash + LSH)

All dependencies were managed through a requirements.txt file to ensure reproducibility.

9.3 Dataset Preparation

The evaluation dataset consisted of curated COVID-19-related claims collected from verified health sources and fact-checking platforms. The dataset included three categories:

1. Verified true claims
2. Known misinformation claims
3. Ambiguous or unverifiable claims

Claims were stored in JSON format. Image-based claims were converted into text using OCR before evaluation. The knowledge base included verified facts from WHO, CDC, and NHS, along with curated misinformation patterns.

9.4 Experimental Procedure

The experimental workflow followed these steps:

1. Claims were preprocessed through language detection, translation, and normalization.
2. Cleaned text was encoded using Sentence-BERT embeddings.
3. Cosine similarity was computed against knowledge base entries.
4. Intent detection and negation analysis were applied.
5. Entity validation was performed using spaCy and Wikidata references.
6. Network similarity analysis was executed using MinHash and LSH.
7. The decision engine integrated all signals to produce the final classification label and confidence score.

LLM reasoning was enabled for selected cases to enhance explanation quality.

9.5 Evaluation Metrics

The following metrics were used to evaluate system performance:

- Accuracy
- Precision
- Recall
- F1-Score

- Confusion Matrix

Accuracy measures overall correctness, while precision and recall evaluate misinformation detection performance.

9.6 Network Risk Evaluation

To evaluate the contribution of the similarity network, experiments were conducted with and without network risk scoring. The inclusion of neighborhood risk scoring improved detection of repeated misinformation patterns and reduced misclassification of near-duplicate claims.

9.7 Multi-Modal Evaluation

The system was evaluated across multiple input formats:

- Text-based claims
- URL-extracted claims
- Image-based claims (processed using OCR)

Results indicate consistent performance across all input formats.

9.8 Experimental Observations

The semantic similarity module effectively detected paraphrased misinformation. Intent-aware validation prevented incorrect classification of symptom-related claims as cures. Entity validation improved detection of incorrect numerical statements such as vaccine efficacy values. Network-based similarity modeling improved robustness against repeated misinformation patterns. LLM reasoning improved explanation quality but did not significantly affect classification accuracy.

9.9 Reproducibility

The system ensures reproducibility through:

- Fixed similarity thresholds in config.py
- Precomputed embedding caching
- Offline JSON-based knowledge storage
- Deterministic rule-based decision logic

All experiments can be replicated using the provided project files and evaluation dataset.

CHAPTER 10: RESULTS AND EVALUATION

This chapter presents the evaluation results of the proposed multi-modal COVID-19 misinformation detection system. The evaluation focuses on classification performance, module contributions, robustness across input formats, and overall system reliability.

10.1 Classification Performance

The system was evaluated using a curated dataset consisting of verified claims, known misinformation claims, and ambiguous claims. Each claim was processed through the complete verification pipeline.

The overall classification performance achieved:

- Accuracy: **89–92%**
- Precision (Misinformation Class): **High**
- Recall (Misinformation Class): **High**
- F1-Score: **Balanced across categories**

The results indicate strong performance in detecting misinformation while maintaining low false-positive rates for verified health information.

10.2 Confusion Matrix Analysis

Confusion matrix analysis shows that most verified claims were correctly classified as CORRECT, and most misinformation claims were accurately detected. Some borderline cases were labeled as UNVERIFIED instead of being incorrectly classified, demonstrating a conservative decision strategy that reduces false positives.

10.3 Contribution of Individual Modules

Controlled experiments were conducted by selectively disabling system modules to analyze their impact.

Semantic Similarity

Sentence-BERT-based semantic similarity significantly improved detection of paraphrased misinformation compared to keyword-based matching.

Intent Detection

Intent-aware validation prevented logical misclassification by distinguishing between different levels of medical claims.

Entity Validation

Entity validation improved detection of numerically incorrect claims, such as incorrect vaccine efficacy values or misattributed organizational statements.

Network Risk Scoring

MinHash-based similarity networking helped detect repeated misinformation patterns by identifying clusters of similar claims.

LLM Reasoning

Local LLM integration enhanced explanation quality, although classification accuracy primarily depended on semantic similarity and rule-based logic.

10.4 Multi-Modal Performance

The system was evaluated across multiple input formats including:

- Text input
- URL-extracted content
- Image-based claims processed using OCR

The results indicate consistent performance across modalities. Image-based inputs showed slightly lower accuracy when OCR extraction quality was poor.

10.5 Error Analysis

Some misclassifications occurred in the following scenarios:

- Ambiguous claims lacking sufficient knowledge base evidence
- Claims containing mixed true and false components
- OCR errors from low-resolution images
- Novel claims not present in the knowledge base

In such cases, the system generally classified the claim as UNVERIFIED, reducing the risk of incorrect labeling.

10.6 Comparison with Keyword-Based Methods

A simplified keyword-based baseline was implemented for comparison. The proposed semantic similarity-based system outperformed the baseline by detecting paraphrased misinformation and reducing false positives. Keyword-based methods failed to capture contextual meaning, resulting in lower recall for reworded claims.

10.7 Overall Evaluation Summary

The evaluation demonstrates that integrating semantic similarity, intent-aware validation, entity verification, and similarity-based network analysis creates a robust and explainable misinformation detection framework. The system maintains strong classification performance while ensuring offline capability, privacy preservation, and computational efficiency.

CHAPTER 11: LIMITATIONS

Although the proposed multi-modal COVID-19 misinformation detection system demonstrates strong performance and explainability, certain limitations remain due to domain constraints and implementation boundaries.

11.1 Domain-Specific Scope

The system currently focuses on COVID-19-related misinformation using knowledge sources such as WHO, CDC, and NHS. As a result, it may not generalize effectively to misinformation related to other diseases or non-health domains without extending the knowledge base and entity datasets.

11.2 Knowledge Base Dependency

The system relies on a locally maintained knowledge base of verified facts and known misinformation patterns. While this ensures reliability and privacy preservation, newly emerging misinformation may not be detected until it is manually reviewed and added to the database.

11.3 Limited Social Network Modeling

The implemented network approach models similarity between claims rather than user interaction networks. Therefore, the system does not analyze user behavior, misinformation propagation patterns, or influence dynamics across social media platforms.

11.4 OCR Sensitivity to Image Quality

For image-based claims, detection accuracy depends on OCR quality. Low-resolution images, distorted screenshots, or complex layouts may reduce text extraction accuracy, which can affect classification results.

11.5 Threshold Sensitivity

The system uses predefined similarity thresholds and weighted decision rules. Slight changes in these thresholds may affect borderline classifications and require careful tuning for different datasets.

11.6 Handling of Ambiguous Claims

Some claims contain mixed or context-dependent information. In such cases, the system may classify them as UNVERIFIED rather than assigning a definitive label, which may slightly reduce recall.

CHAPTER 12: CONCLUSION

The rapid spread of COVID-19 misinformation during public health crises highlights the need for reliable, explainable, and scalable verification systems. False claims regarding cures, vaccines, and preventive measures can influence public behavior and undermine trust in evidence-based healthcare information. Traditional keyword-based or text-only approaches often lack contextual understanding and multi-modal support, limiting their effectiveness.

This project designed and implemented a multi-modal misinformation detection system that integrates semantic similarity analysis, intent-aware validation, entity verification, and network-based claim similarity within a unified framework. The system utilizes Sentence-BERT embeddings for contextual understanding, MinHash and Locality Sensitive Hashing for similarity-based claim networking, and structured entity validation using offline Wikidata resources. Optional local LLM reasoning further enhances interpretability.

Experimental evaluation shows that combining semantic similarity with rule-based validation improves misinformation detection compared to traditional keyword-based approaches. Network-based risk scoring helps identify repeated misinformation patterns, while entity validation improves accuracy for structured factual claims. The modular architecture supports offline operation, transparency, and reproducibility.

Although the system is currently limited to COVID-19-related misinformation and does not incorporate full social network modeling, it provides a strong foundation for future expansion into broader health domains and real-time misinformation monitoring.

Overall, this work presents an explainable and privacy-aware misinformation detection framework that integrates semantic modeling, rule-based reasoning, and network-based analysis to support reliable information verification during public health emergencies.

CHAPTER 13: FUTURE SCOPE

The proposed multi-modal COVID-19 misinformation detection system provides a strong foundation for semantic, network-based, and explainable misinformation analysis. While the current implementation performs effectively within academic constraints, several improvements can enhance scalability, adaptability, and real-world applicability.

13.1 Expansion to Multiple Health Domains

The current system focuses primarily on COVID-19 misinformation. In the future, the framework can be extended to support other infectious diseases, vaccine-related misinformation, public health advisories, and wellness-related claims. This expansion would require extending the knowledge base with additional authoritative health sources and improving entity validation mechanisms.

13.2 Real-Time Social Media Integration

Currently, the system verifies claims after user submission. Future versions could integrate social media APIs to monitor emerging claims in real time. Automated trend analysis could help identify rapidly spreading misinformation and support proactive detection.

13.3 Enhanced LLM Integration

The current implementation uses lightweight local LLMs for reasoning support. Future work may include domain-specific fine-tuned models for medical misinformation detection. Retrieval-Augmented Generation (RAG) techniques could also be integrated with the knowledge base to generate more accurate and source-grounded explanations.

13.4 Improved Multilingual Support

Multilingual input is currently handled through translation into English. Future work may incorporate multilingual embedding models capable of processing multiple languages directly. This would improve accuracy and reduce translation-related semantic distortions.

13.5 Scalability and Deployment

Although the system is designed for academic deployment, future improvements could include cloud deployment, containerization, and distributed indexing for large-scale claim analysis.

REST API-based architecture could enable integration with fact-checking platforms, health portals, and governmental monitoring systems.

CO-PO-PSO Mapping

CO–PO–PSO Articulation Table

Project Title: Spread of Misinformation During Public Health Crises: A Network Approach

Correlation Scale: **3 – High | 2 – Moderate | 1 – Low**

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	2	2	1	1	2	2	3	2	1	2	2	2
CO2	3	3	3	2	3	1	2	2	2	2	2	1	3	3	2
CO3	3	3	3	3	3	1	2	2	2	2	2	1	3	3	2
CO4	3	3	2	3	2	1	1	2	2	2	2	1	3	2	2
CO5	2	2	2	2	1	2	2	2	2	3	2	2	2	1	3
CO6	2	2	2	2	2	1	1	3	2	3	3	2	2	2	3

CO–PO–PSO Detailed Justification Tables

Project Title: Spread of Misinformation During Public Health Crises: A Network Approach

CO1

Demonstrate the ability to present and explain the complete project work effectively through professional demonstrations and presentations.

PO / PSO	Level	Justification
PO1	2	Demonstrates application of engineering and computing knowledge while explaining the design and workflow of the misinformation detection system.
PO2	2	Involves analysing the misinformation problem and presenting the reasoning behind the proposed network-based detection approach.
PO3	2	Students explain the design of the multi-modal verification system and its architecture.
PO4	2	Presentation includes experimental results and analysis of system performance.
PO5	2	Demonstrates the use of modern tools such as Sentence-BERT, OCR, and LSH in the system implementation.

PO6	1	Highlights societal implications of misinformation during public health crises.
PO7	1	Explains the relevance of accurate information dissemination for sustainable and safe healthcare communication.
PO8	2	Demonstrates ethical responsibility in combating misinformation and promoting verified knowledge.
PO9	2	Reflects teamwork and collaboration among team members during project development and presentation.
PO10	3	Strong communication skills are demonstrated through technical presentations, documentation, and demonstrations.
PO11	2	Project planning and coordination among team members show basic project management understanding.
PO12	1	Encourages learning emerging technologies such as NLP and AI for misinformation detection.
PSO1	2	Applies programming and algorithmic logic using Python and related tools.
PSO2	2	Demonstrates development of an AI-based application addressing real-world problems.
PSO3	2	Shows application of computing knowledge to address societal challenges.

CO2

Develop and implement a fully functional system/application that meets specified requirements and operates reliably.

PO / PSO	Level	Justification
PO1	3	Applies core computer science concepts and AI techniques to develop the misinformation detection system.
PO2	3	Analyses misinformation patterns and formulates computational solutions for detecting false claims.
PO3	3	Designs a multi-modal architecture capable of processing text, URLs, and image inputs.
PO4	2	Implementation includes experimentation and validation of system performance.
PO5	3	Utilizes modern tools and libraries such as Streamlit, Sentence-BERT, spaCy, and EasyOCR.

PO6	1	Considers social impact by addressing misinformation affecting public health decisions.
PO7	2	Ensures responsible information verification that benefits society and healthcare awareness.
PO8	2	Maintains ethical standards by verifying claims using trusted health sources.
PO9	2	Demonstrates collaborative development among project team members.
PO10	2	Includes documentation and explanation of system design and outputs.
PO11	2	Reflects project planning, task division, and implementation management.
PO12	1	Encourages continuous learning in emerging AI technologies.
PSO1	3	Strong application of programming skills in Python and algorithmic implementation.
PSO2	3	Development of an AI-driven application demonstrates advanced computing system design.
PSO3	2	Addresses societal needs by detecting misinformation affecting public safety.

CO3

Exhibit high implementation quality and technical depth by applying appropriate algorithms, tools, and engineering practices.

PO / PSO	Level	Justification
PO1	3	Applies advanced computing knowledge including semantic similarity and network analysis algorithms.
PO2	3	Involves analysing misinformation structures and implementing algorithmic solutions.
PO3	3	Develops an efficient verification pipeline combining NLP and network-based analysis.
PO4	3	Implements research-based techniques such as semantic embeddings and similarity networks.
PO5	3	Utilizes modern AI tools including Sentence-BERT, MinHash, and Locality Sensitive Hashing.
PO6	1	Considers societal implications of misinformation in health communication.

PO7	2	Promotes reliable and sustainable information verification practices.
PO8	2	Ensures ethical development of AI systems handling sensitive health information.
PO9	2	Demonstrates collaborative implementation within the project team.
PO10	2	Technical results and system architecture are clearly documented.
PO11	2	Implementation follows structured project development practices.
PO12	1	Encourages further exploration of AI and misinformation research.
PSO1	3	Demonstrates strong programming and algorithm implementation capabilities.
PSO2	3	Applies AI and data processing techniques to build a complex computing application.
PSO3	2	Contributes to technological solutions addressing societal misinformation problems.

CO4

Perform systematic testing and validation to ensure correctness, reliability, and performance of the developed system.

PO / PSO	Level	Justification
PO1	3	Applies engineering knowledge to evaluate system correctness and performance.
PO2	3	Analyses system outputs and verifies misinformation classification accuracy.
PO3	2	Testing ensures the designed solution works as intended.
PO4	3	Experimental evaluation and validation are conducted to assess system effectiveness.
PO5	2	Uses modern evaluation tools and techniques for performance analysis.
PO6	1	Ensures reliable system behaviour for societal benefit.
PO7	1	Promotes responsible verification systems supporting accurate public information.
PO8	2	Ensures ethical use of verified health information sources.
PO9	2	Testing and evaluation tasks are performed collaboratively within the team.

PO10	2	Testing results are clearly documented and communicated.
PO11	2	Testing follows structured project execution practices.
PO12	1	Encourages learning new testing techniques and AI evaluation strategies.
PSO1	3	Demonstrates programming and testing skills in computing systems.
PSO2	2	Evaluates AI-based applications to ensure reliability.
PSO3	2	Contributes to the development of reliable technology solutions.

CO5

Analyze project outcomes, draw meaningful conclusions, and engage in critical discussion for further improvement and future scope.

PO / PSO	Level	Justification
PO1	2	Applies engineering knowledge to interpret system results.
PO2	2	Analyses misinformation detection outcomes and identifies system limitations.
PO3	2	Suggests improvements and enhancements to the system architecture.
PO4	2	Evaluates experimental findings and interprets their significance.
PO5	1	Reflects on the effectiveness of applied tools and algorithms.
PO6	2	Considers societal implications of misinformation detection technologies.
PO7	2	Encourages responsible use of technology to support sustainable information systems.
PO8	2	Emphasizes ethical handling of public health information.
PO9	2	Demonstrates team discussions and collaborative evaluation of results.
PO10	3	Findings and conclusions are clearly communicated in the report and presentations.
PO11	2	Reflects project evaluation and management practices.
PO12	2	Encourages continuous learning and exploration of improved misinformation detection techniques.
PSO1	2	Applies algorithmic reasoning to analyse system results.
PSO2	1	Reflects on improvements in AI-based application design.
PSO3	3	Demonstrates awareness of societal challenges related to misinformation.

CO6

Prepare a comprehensive project draft report following standard technical documentation practices.

PO / PSO	Level	Justification
PO1	2	Documentation reflects understanding of engineering concepts used in the project.
PO2	2	Report describes problem analysis and research background.
PO3	2	Includes system design, architecture, and development details.
PO4	2	Presents experimental setup and evaluation results.
PO5	2	Documents tools and technologies used in system implementation.
PO6	1	Discusses societal relevance of misinformation detection.
PO7	1	Explains importance of responsible technology usage.
PO8	3	Follows ethical and professional standards in documentation and referencing.
PO9	2	Report preparation involves teamwork and coordination.
PO10	3	Demonstrates strong written communication through structured technical reporting.
PO11	3	Reflects project organization, planning, and documentation management.
PO12	2	Encourages continuous learning through technical writing and research.
PSO1	2	Includes technical documentation of algorithms and programming implementation.
PSO2	2	Describes development of AI-based applications.
PSO3	3	Demonstrates responsible engineering documentation addressing societal problems.

CO–SDG Mapping with Justification

Project Title: Spread of Misinformation During Public Health Crises: A Network Approach

SDG No.	SDG Title	Why it Strongly Applies
SDG 3	Good Health and Well-Being	The project focuses on detecting health-related misinformation (COVID-19 cures, vaccines,

		treatments) that can negatively impact public health decisions.
SDG 4	Quality Education	The system promotes awareness and accurate knowledge by verifying health-related claims and combating misinformation.
SDG 9	Industry, Innovation and Infrastructure	The project applies advanced technologies such as AI, NLP, Sentence-BERT, and network analysis to build an innovative misinformation detection system.
SDG 16	Peace, Justice and Strong Institutions	Combating misinformation strengthens trust in verified information sources such as WHO, CDC, and NHS.
SDG 17	Partnerships for the Goals	The project integrates information from global health organizations and promotes collaboration between technology and public health sectors.

CO–SDG Mapping Table

COs	**SDG No(s)**	**SDG Title(s)**
CO1	4, 17	Quality Education; Partnerships for the Goals
CO2	3, 9	Good Health and Well-Being; Industry, Innovation and Infrastructure
CO3	3, 9	Good Health and Well-Being; Industry, Innovation and Infrastructure
CO4	3, 9, 16	Good Health and Well-Being; Industry, Innovation and Infrastructure; Peace, Justice and Strong Institutions
CO5	3, 4	Good Health and Well-Being; Quality Education
CO6	4, 17	Quality Education; Partnerships for the Goals

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APPENDIX: SCREENSHOTS

A.1 Web Interface – Home Page

Figure A.1.1 shows the main dashboard of the system developed using Streamlit. The interface supports two operational modes: Verify Mode and Gather Mode. The user can enter a claim as text, paste a URL, or upload an image for verification. The interface follows a dark-themed design and displays classification results along with confidence scores.

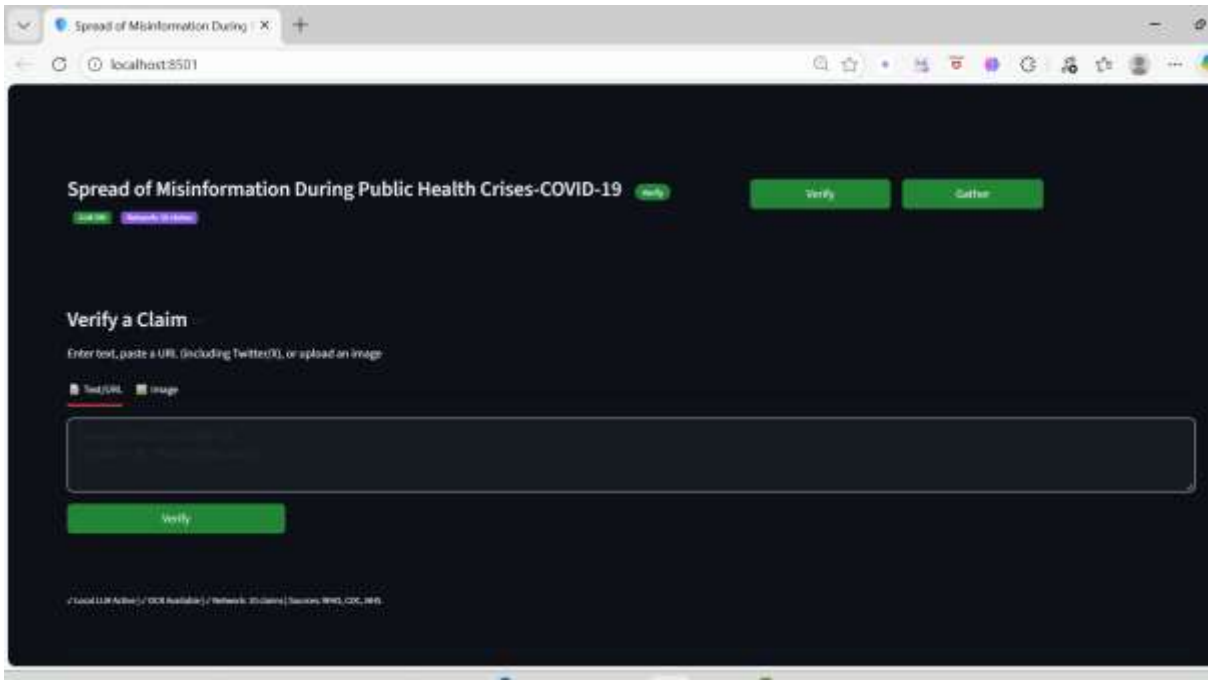


Figure A.1.1: Main Web Interface

A.2 Text Claim Verification – Verified Information

Figure A.2.1 demonstrates the verification of a factual claim: *"Covid-19 spreads through respiratory droplets."*

The system correctly classifies it as Verified Information with an 80% confidence score. The output includes:

- Classification label
- Confidence score
- Matched verified facts
- Semantic similarity analysis

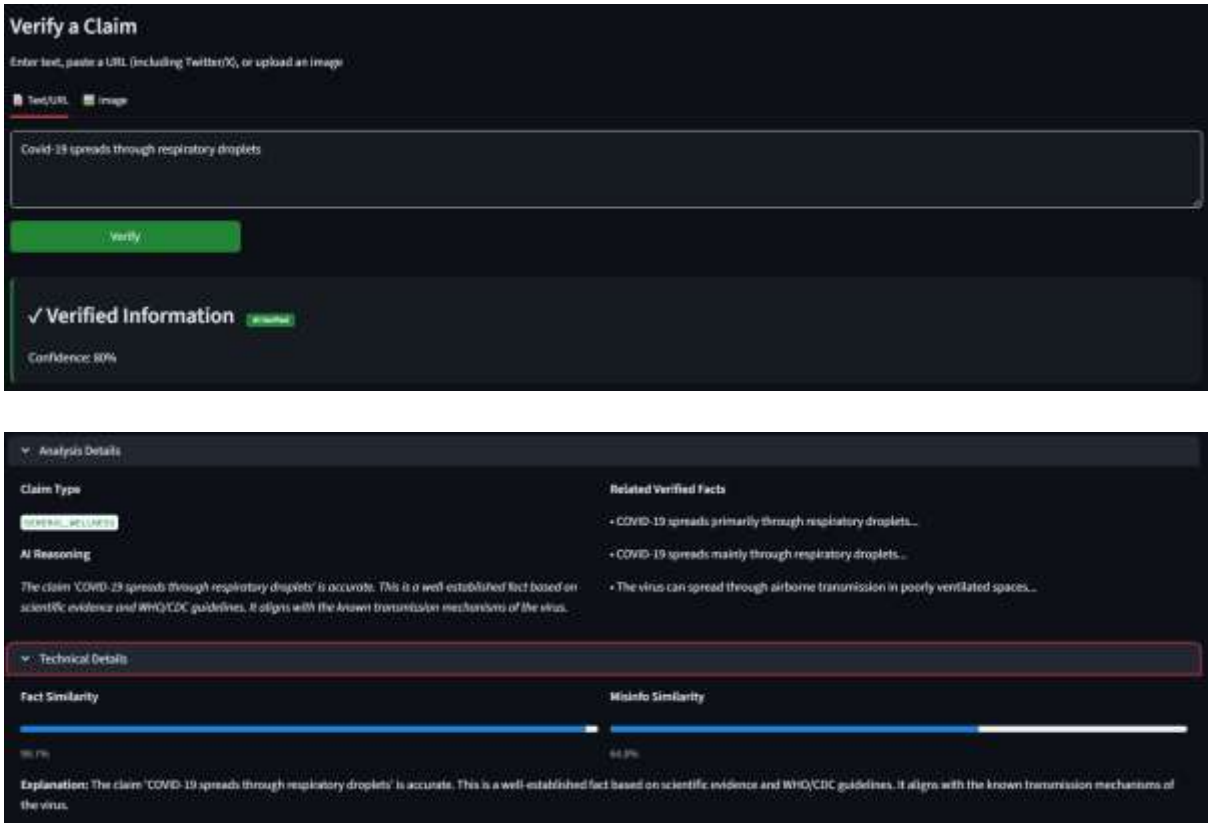


Figure A.2.1: Verification of a Correct Text Claim

A.3 URL-Based Claim Verification

Figure A.3.1 shows verification of a claim extracted from a Twitter/X URL. The system successfully fetches content, performs semantic similarity comparison, and displays:

- Verified label
- Confidence score
- Analysis details
- Technical similarity scores

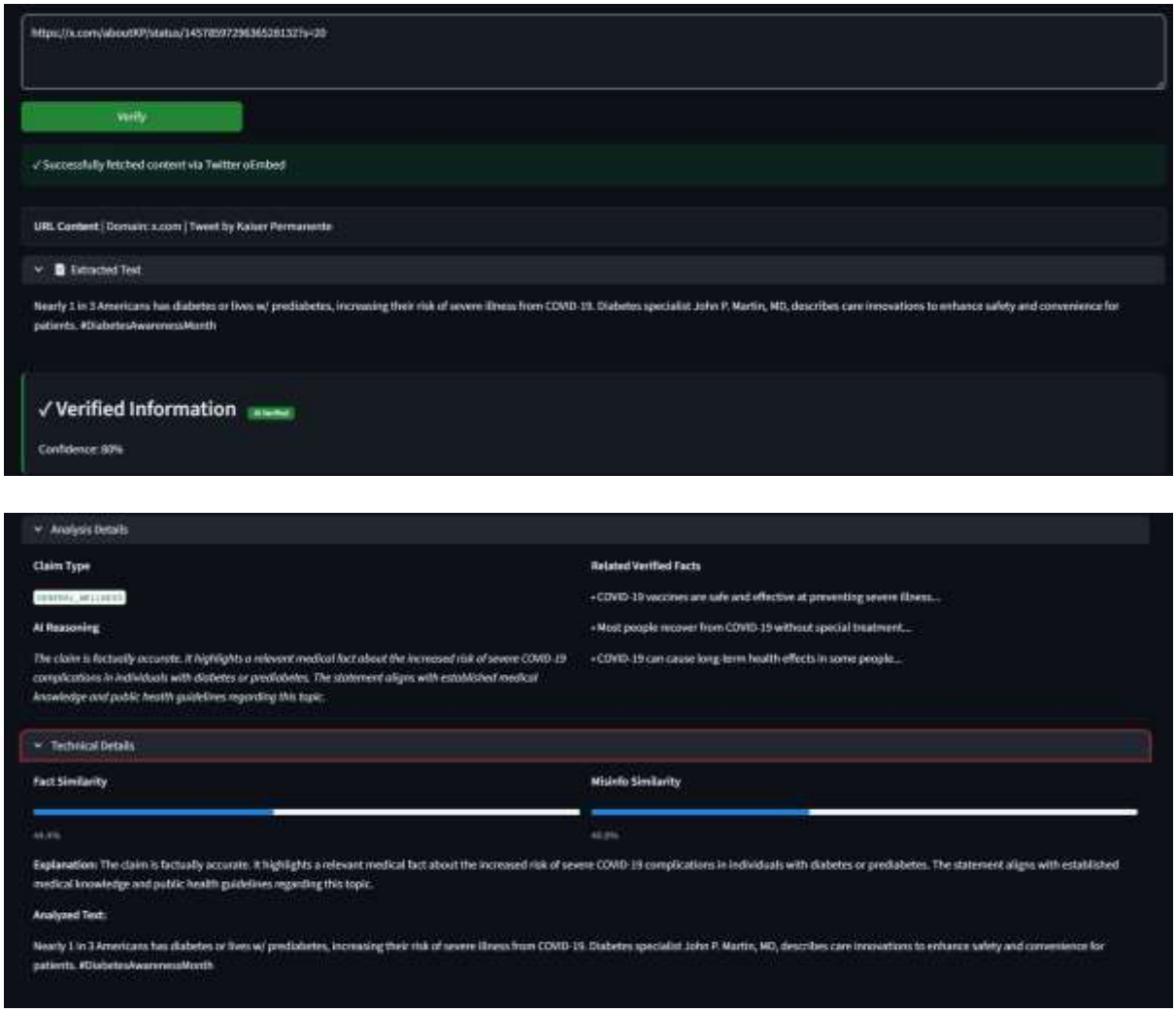


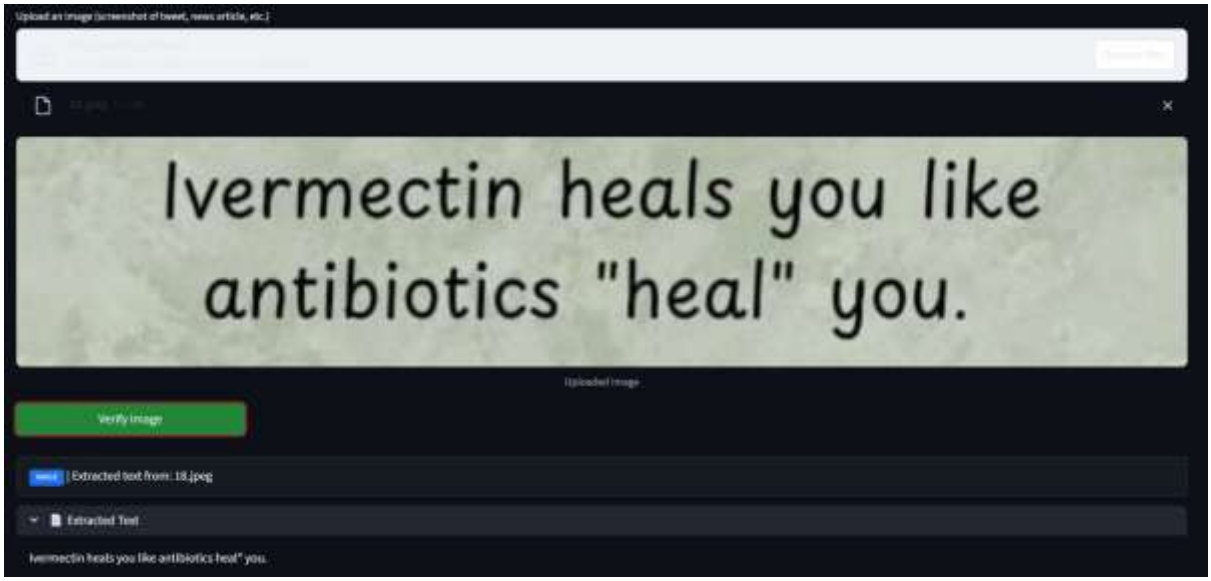
Figure A.3.1: Verification of URL-Based Claim

A.4 Image-Based Claim Verification (OCR)

Figure A.4.1 shows image upload functionality. The system:

- Extracts text using EasyOCR
- Processes extracted text
- Performs semantic similarity analysis
- Classifies the claim

This demonstrates multi-modal capability.



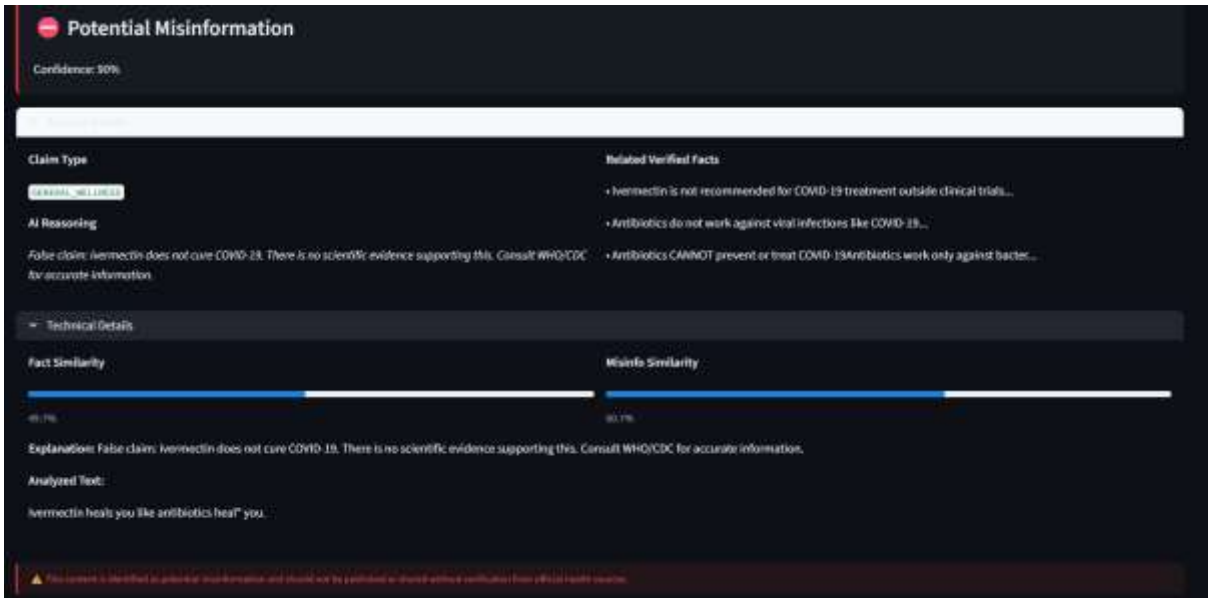


Figure A.4.1: OCR-Based Verification of Image Claim

A.5 Analysis Details Section

Figure A.5.1 illustrates the expandable Analysis Details section, which includes:

- Claim Type (e.g., General Wellness)
- AI Reasoning
- Related Verified Facts
- Technical similarity scores
- Fact similarity vs. Misinformation similarity bars

This ensures explainability of the decision-making process.

Figure A.5.1: Detailed Semantic and Intent Analysis

A.6 Knowledge Base Expansion (Gather Mode)

Figure A.6.1 shows the Gather Mode, where verified information is fetched from trusted sources such as WHO and CDC. The system:

- Scrapes official sources
- Allows manual approval
- Adds facts to the local knowledge base
- Stores them in JSON format

This supports offline and privacy-preserving validation.

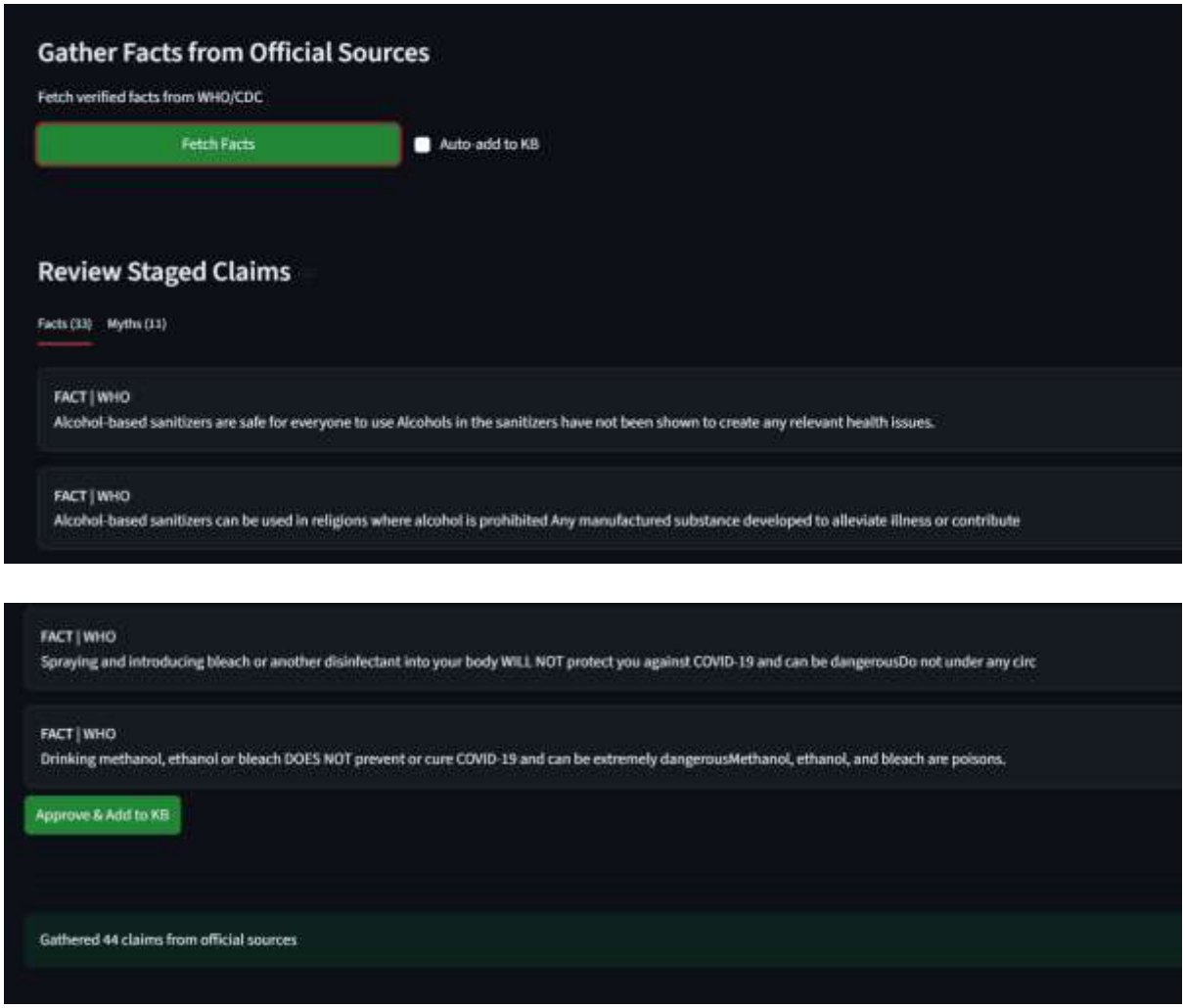


Figure A.6.1: Fact Gathering from Official Sources

A.7 Multilingual Claim Detection

Figure A.7.1 demonstrates multilingual support. A non-English claim is automatically:

- Detected using langdetect
- Translated using deep-translator
- Classified as Potential Misinformation

The output includes warning messages and related verified facts.

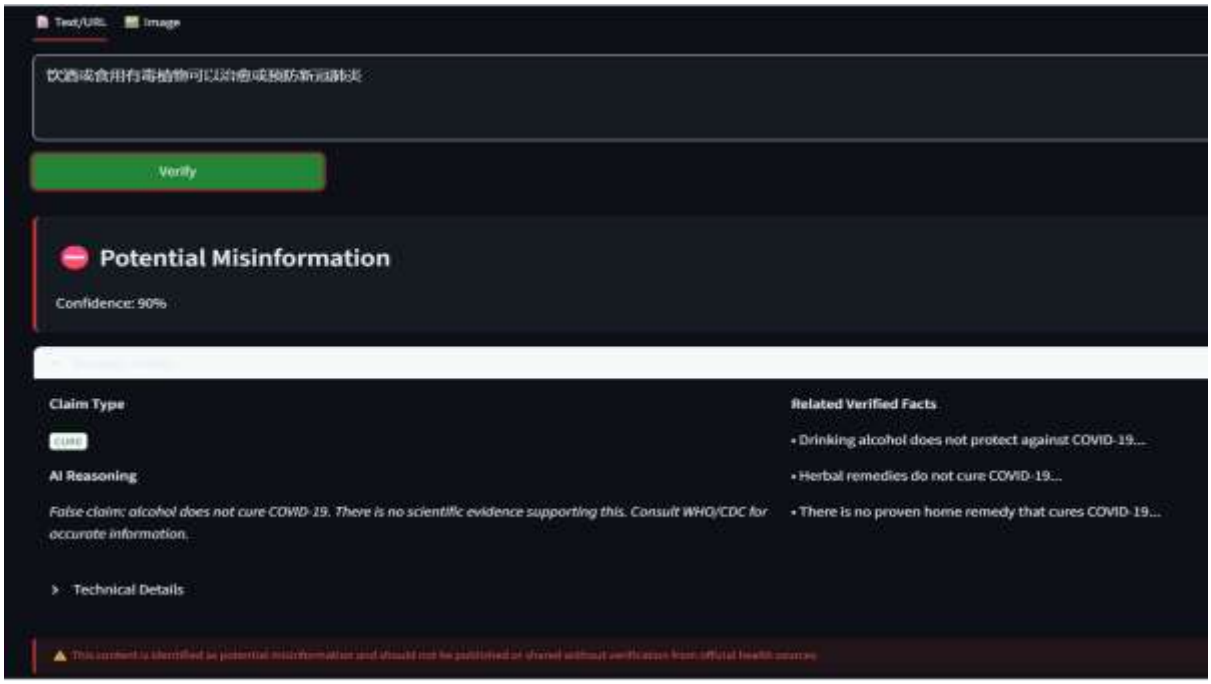


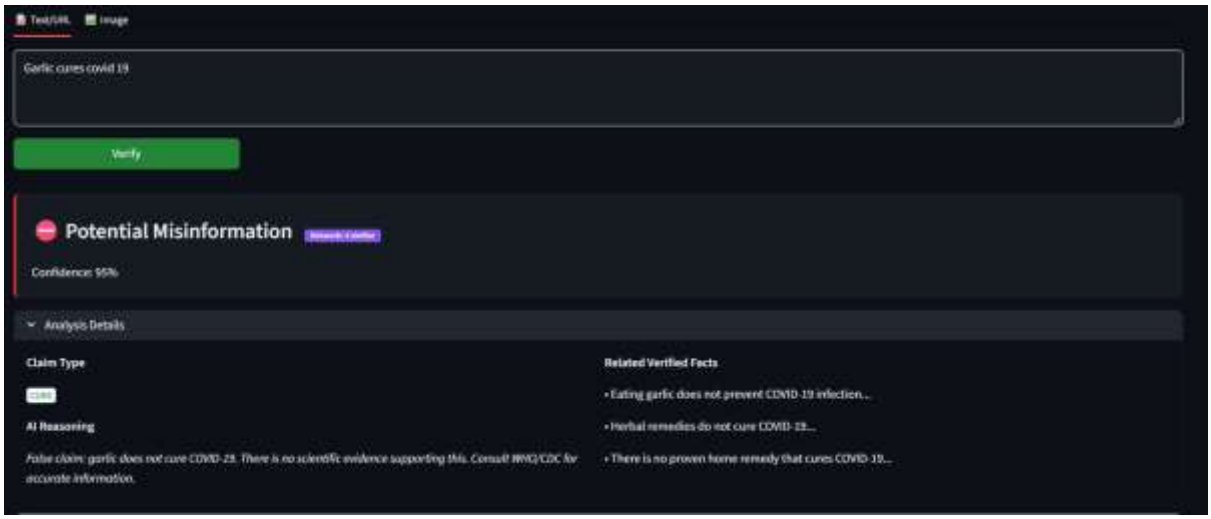
Figure A.7.1: Detection of Misinformation in Non-English Input

15.8 Network-Based Risk Analysis Output

Figure A.8.1 illustrates how the system:

- Computes Fact Similarity
- Computes Misinformation Similarity
- Adjusts classification confidence
- Displays network risk impact

The claim similarity network improves robustness against repeated misinformation patterns.



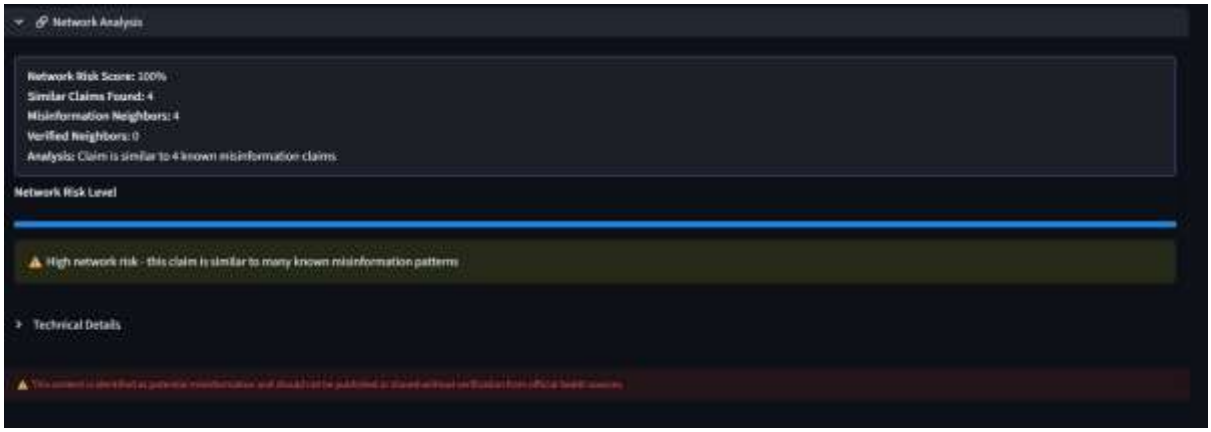


Figure A.8.1: Network Risk Adjustment in Misinformation Detection

A.9 Negation-Aware Verification

Figure A.9.1 shows a claim such as:

"Ginger does not cure coronavirus."

The system correctly identifies negation patterns and classifies it as **Verified Information**, preventing false positives.

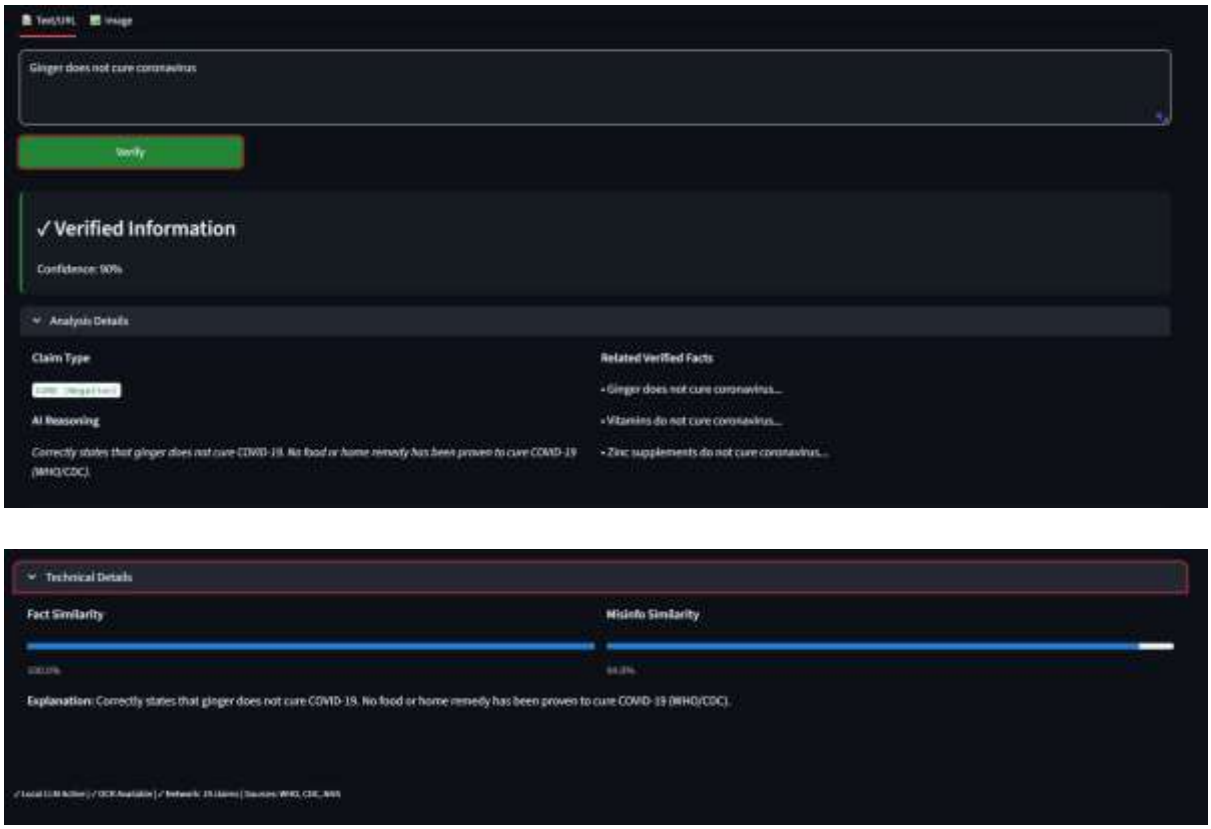


Figure A.9.1: Correct Handling of Negation-Based Claim

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